

Nucleation density programmes the viscoelastic gel point of a percolation network

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Passive microrheology locates the viscoelastic gel point through the Winter–Chambon condition $\alpha = 0.5$. Using three-dimensional random walks in the void of site-percolation lattices, we identify this gel point with the void-percolation transition, $p'_c \simeq 1 - p_c$, and show that a single nucleation-density parameter κ tunes it continuously from 0.68 towards unity. Finite-size scaling yields an effective exponent $\nu \approx 1.6$ with no systematic dependence on κ for $\kappa \leq 0.8$, distinct from static percolation, indicating that nucleation density shifts the gel point while preserving its critical character.

Gelation—the transition of a polymer solution or colloidal suspension from a viscous fluid to a viscoelastic solid—is characterised by a gel point p'_c at which the loss tangent $\tan \delta = G''/G'$ becomes frequency independent and equal to unity [1, 2]. In the Winter–Chambon picture this is the phase angle $\delta = 45^\circ$ and a power-law mean squared displacement (MSD) $\langle r^2(t) \rangle \propto t^{0.5}$, both accessible by passive microrheology through the Generalised Stokes–Einstein Relation (GSER) [3–5]. The gel point is routinely treated as a fixed property of composition—set by concentration, cross-link density, and temperature [6, 7]. What is rarely asked is whether it also depends on *how* the network forms: on the spatial growth dynamics of the cluster itself, which in real colloidal and biopolymer gels range from open fractal aggregates to compact space-filling domains [8–10].

Microrheology has previously been used to locate the gel point of real gels [11, 12], and growth or nucleation rules are known to shift the percolation threshold and even change the order of the transition [13–16]. What has been missing is a single picture in which the passive-microrheology gel point is *identified* with the void-percolation transition and then moved continuously by one morphological parameter—which is what the nucleation density κ provides here.

Here we show, by direct simulation, that the gel point is set by growth dynamics and can be programmed continuously through a single nucleation-density parameter, while its effective critical character is preserved. The approach is grounded in how microrheology actually works: probe particles diffuse through the *void* (fluid) phase of the network, not through the network [17–19]. We therefore place random walkers on the unoccupied sublattice of an $L \times L \times L$ simple cubic lattice, sweep the occupation probability p , and extract the anomalous-diffusion exponent $\alpha(p)$ from $\langle r^2(t) \rangle \propto t^\alpha$; we identify the gel point operationally as the value of p at which $\alpha = 0.5$, equivalently $\delta = 45^\circ$ for a power-law MSD through the GSER [1, 2, 20]. Walkers move to a uniformly chosen nearest neighbour if it is unoccupied and otherwise dwell for a short binding time; the ensemble MSD is accumulated online and α is fitted over the intermediate decade $[L_W/100, L_W/10]$ (see Supplemental Material [21] for all

methods).

Network growth is controlled by a nucleation-density parameter $\kappa \in [0, 1]$. At each increment in p , a fraction $(1 - \kappa)$ of the newly occupied sites are placed as independent random nuclei and the remainder are grown by nearest-neighbour adjacency from the existing cluster frontier. Thus $\kappa = 0$ is classical Bernoulli site percolation (every site an independent seed: maximal nucleation density), $\kappa = 1$ is single-seed compact (Eden [22]) growth, and intermediate κ interpolates between a spatially uniform obstacle field and a compact, clustered network. The seed fraction $1 - \kappa$ is the nucleation density—*independent nuclei per newly occupied site*—so κ indexes that density inversely. We treat κ as a coarse-grained proxy for nucleation conditions that are varied experimentally through cross-linker concentration, thermal-ramp rate, or ionic strength, rather than a parameter set directly in the laboratory.

The gel point is the void-percolation transition.—For random growth ($\kappa = 0$), $\alpha(p)$ falls from Fickian diffusion ($\alpha \approx 1$) through the gel criterion and arrests sharply [Fig. 1]. The crossing is essentially size independent above $L = 200$, extrapolating to

$$p'_c(\infty) \approx 0.681, \quad (1)$$

within one percent of the complementary void-percolation threshold $1 - p_c = 0.6884$ ($p_c = 0.3116$). The identification is geometric: the walkers occupy the void sublattice, itself a site-percolation problem at occupation $1 - p$, which loses spanning connectivity at $p = 1 - p_c$. A small residual offset is expected, because p'_c is defined dynamically through α rather than a static spanning condition; at the incipient void cluster the MSD is $\langle r^2(t) \rangle \propto t^{0.5}$ —the Winter–Chambon criterion. The rheological gel point of a randomly grown network is therefore not an independent parameter but is fixed by the geometry of its void phase; moving it requires changing the network morphology.

Nucleation density tunes the gel point.—Increasing κ (lowering the nucleation density) shifts the $\alpha = 0.5$ crossing monotonically to higher p [Fig. 2]. A compact, connected cluster of a given mass occupies a more localised

region than the same number of randomly placed sites, leaving a larger and better-connected void that percolates only at higher occupation. Finite-size scaling gives thermodynamic gel points rising smoothly with κ : $p'_c(\infty) = 0.681, 0.702, 0.722$ for $\kappa = 0, 0.6, 0.8$, resolved cleanly, rising to $p'_c(\infty) \approx 0.79\text{--}0.80$ at $\kappa = 0.97$ —a broadened crossover point carrying larger uncertainty (see below)—and continuing towards the single-seed Eden limit, where a single compact cluster can fill the lattice before fragmenting the void ($p'_c \rightarrow 1$, treated as an asymptotic bound rather than a finite critical point). A fine κ -grid at fixed L resolves the full curve $p'_c(\kappa)$ [21], monotonic with two regimes separated near a $\sim 1\%$ seed fraction ($\kappa \approx 0.99$): a gradual rise where many nuclei keep the morphology quasi-random, and a steep rise towards unity where single-cluster growth dominates. Across the accessible range the gel point spans $p'_c \approx 0.68\text{--}0.91$, approaching unity only asymptotically in the Eden limit.

One universality class.—Does moving p'_c with κ change the character of the transition? Finite-size scaling across $L = 50\text{--}750$ answers no. The gel point converges by $L \approx 200$, so the shift $p'_c(L) = p'_c(\infty) + aL^{-1/\nu}$ fixes $p'_c(\infty)$ robustly but leaves ν underdetermined; the exponent is instead obtained from two estimators with genuine leverage—the scaling of the transition width $w(L) \sim L^{-1/\nu}$ and the data collapse of $\alpha(p, L)$ [Fig. 3]. Both give $\nu_{\text{eff}} \approx 1.6$, with jackknife intervals that overlap across $\kappa \leq 0.8$ and show no systematic trend in κ :

$$\nu_{\text{eff}} \approx 1.6 \quad (\text{no systematic } \kappa\text{-trend, } \kappa \leq 0.8), \quad (2)$$

the strongly clustered $\kappa \geq 0.97$ conditions lie in the Eden crossover, where the transition broadens and the exponent estimators become unreliable, and are excluded from the estimate [21]. This effective exponent differs markedly from the static 3D site-percolation value $\nu = 0.88$. The difference is expected: p'_c is defined *dynamically*, through the probe-diffusion exponent, rather than through a static order parameter, so the width over which α crosses 0.5 need not scale with the static correlation length. The substantive result is the *absence of a systematic κ -trend* in ν_{eff} : within our resolution, nucleation density moves the gel point without changing its effective critical character. The gel point is thus a programmable but non-universal threshold, not a family of distinct critical behaviours.

Structure confirms dynamics.—The identification is not merely dynamical. Computing the void-percolation threshold directly—the loss of a spanning empty cluster on the same lattices—reproduces the microrheological gel point wherever the network is percolation-like: the geometric and dynamical thresholds coincide to within our resolution for $\kappa \lesssim 0.8$ [Fig. 4]. They separate only for $\kappa \gtrsim 0.85$, the dynamical point lagging the structural one as the void fragments into large finite pockets, marking the same Eden crossover seen in ν_{eff} . That a transport exponent and a connectivity threshold agree where the

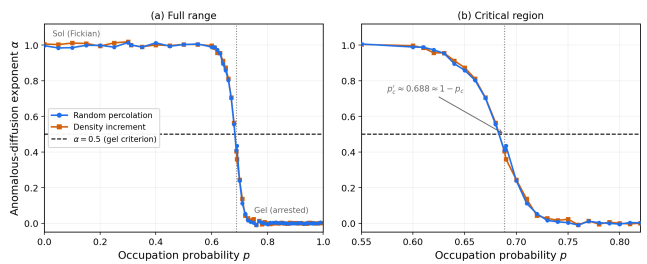


FIG. 1. The gel point is the void-percolation transition. Anomalous-diffusion exponent $\alpha(p)$ for random ($\kappa = 0$) growth: Fickian diffusion ($\alpha \approx 1$) gives way, through the Winter–Chambon criterion $\alpha = 0.5$ (dashed), to arrest. Finite-size scaling places the gel point at $p'_c(\infty) \approx 0.681$, within one percent of the void-percolation threshold $1 - p_c = 0.6884$ ($p_c = 0.3116$).

medium is critical, and part only as it compactifies, is direct evidence that p'_c is the void-percolation transition, dialled by κ .

Finally, the results map directly onto measurable rheology. Because the MSD is a power law over the anomalous window, the GSER yields $G', G'' \propto \omega^\alpha$ with a frequency-independent phase angle $\delta = 90\alpha$; at p'_c this gives $G' \approx G''$ and $\delta \approx 45^\circ$, the Winter–Chambon signature, across the conditions examined [21]. A microrheology experiment on a network whose nucleation density is varied during formation should therefore observe the gel point shift in the direction, and qualitatively by the amount, that $p'_c(\kappa)$ predicts.

These results recast the gel point as a candidate structural design parameter. Rather than a fixed consequence of composition, it is set by the growth dynamics of the network-forming phase and can be dialled across $0.68 \lesssim p'_c \lesssim 0.91$ (with unity as an asymptotic bound) by controlling nucleation density—while, because the effective critical character shows no systematic κ -dependence, the qualitative rheology at the gel point is preserved. The void-percolation baseline $p'_c \simeq 1 - p_c$ supplies the reference; nucleation density supplies the tuning; and finite-size scaling indicates that the tuning is consistent with a single effective critical character. Whether $\nu_{\text{eff}} \approx 1.6$ admits an exact value, and how it relates to the spectral and fractal dimensions of the incipient void cluster, are natural next questions.

[Acknowledgments to be completed.]

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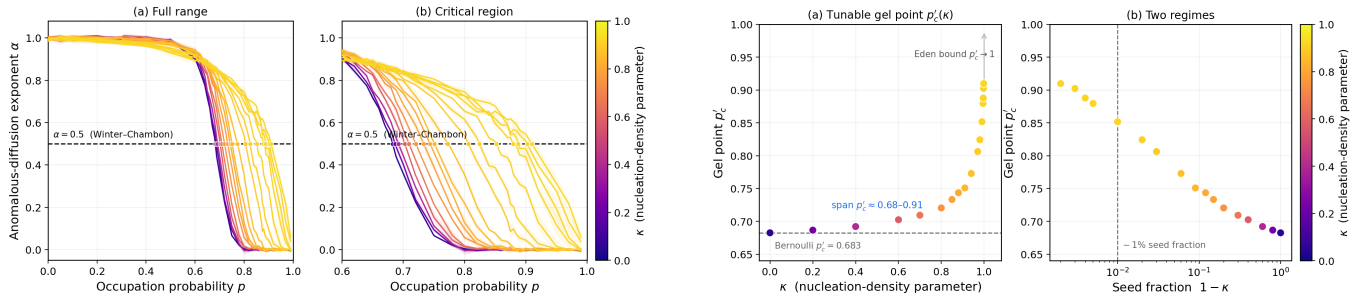


FIG. 2. **Nucleation density tunes the gel point.** (a) $\alpha(p, \kappa)$: the $\alpha = 0.5$ crossing shifts monotonically to higher p as κ increases from 0 (random) towards 1 (single-seed Eden). (b) Gel-point curve $p'_c(\kappa)$, spanning $p'_c \approx 0.68$ – 0.91 over the accessible range, with two regimes separated near a $\sim 1\%$ seed fraction.

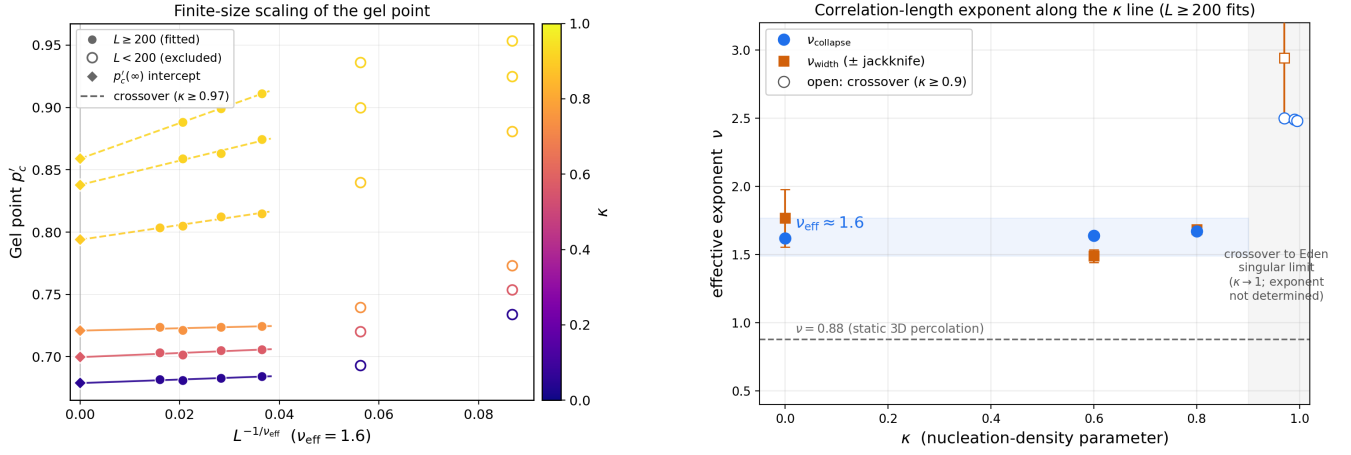


FIG. 3. **A single tunable universality class.** (a) Finite-size scaling $p'_c(L) = p'_c(\infty) + aL^{-1/\nu}$ for each κ (filled: $L \geq 200$ used in fits; open: smaller sizes excluded). (b) Effective exponent $\nu_{\text{eff}}(\kappa)$ from data collapse and transition-width scaling: both agree at $\nu_{\text{eff}} \approx 1.6$, independent of κ and distinct from static 3D percolation ($\nu = 0.88$, dashed). The $\kappa = 0.97$ point (open) lies near the Eden crossover and is excluded from the exponent estimate.

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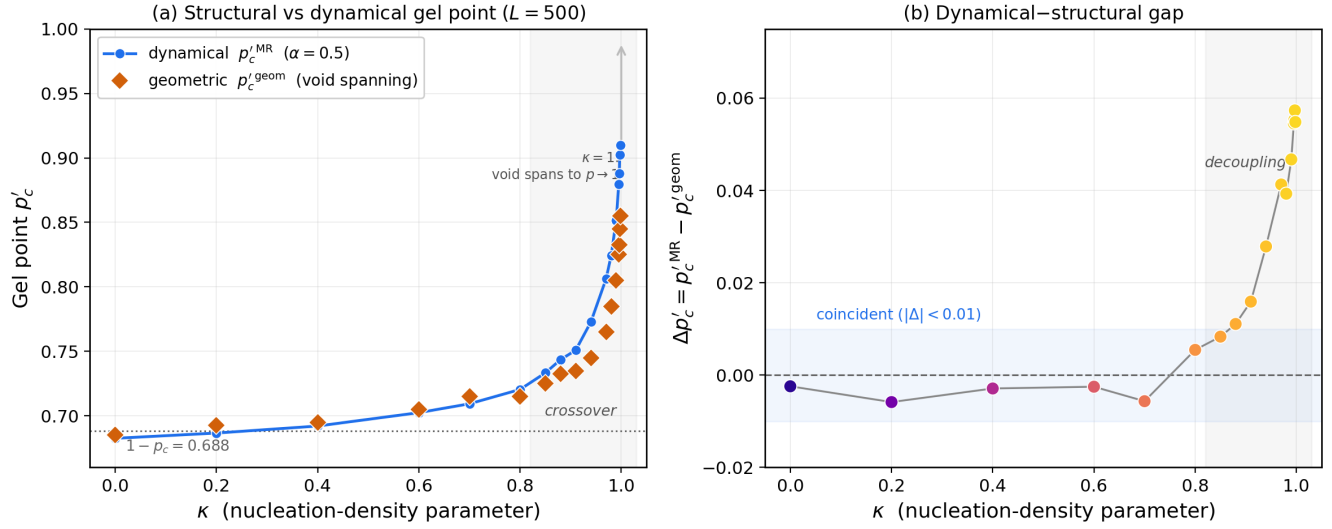


FIG. 4. **The microrheological gel point is the void-percolation threshold.** (a) Dynamical gel point $p'_c{}^{\text{MR}}$ (the $\alpha = 0.5$ crossing) and the geometric void-spanning threshold $p'_c{}^{\text{geom}}$ (loss of a face-to-face spanning empty cluster), both at $L = 500$. (b) Their difference $\Delta p'_c = p'_c{}^{\text{MR}} - p'_c{}^{\text{geom}}$: the two coincide to within the p -grid resolution for $\kappa \lesssim 0.8$ and decouple through the Eden crossover, the dynamical point lagging the structural one. At $\kappa = 1$ the void of a single compact cluster spans until $p \rightarrow 1$ (no finite- p threshold).

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Supplemental Material

Nucleation density programmes the viscoelastic gel point of a percolation network

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This Supplemental Material gives the lattice and random-walk model, the nucleation-density growth algorithm, the finite-size-scaling protocol and its full results, the fixed- L gel-point curve, the frequency-domain (GSER) analysis, robustness checks on the growth rule, and computational details. Equation, figure, table, and section numbers are prefixed “S”.

LATTICE MODEL AND THE NUCLEATION-DENSITY (κ) GROWTH ALGORITHM

We use a simple cubic lattice of side L with periodic boundary conditions. Each site is *occupied* (network phase) or *unoccupied* (void/fluid phase); the occupation probability p sets the target occupied count $N_{\text{target}} = \lfloor pL^3 \rfloor$. Lattices are built cumulatively over the p -sweep, so the occupied set at higher p contains that at lower p as a subset—consistent with progressive gelation and ensuring continuity of $\alpha(p)$.

Growth is governed by a nucleation-density parameter $\kappa \in [0, 1]$. At each increment, of the N_{new} sites to be added,

$$N_{\text{seeds}} = \lfloor (1 - \kappa) N_{\text{new}} \rfloor \quad (\text{S1})$$

are placed at uniformly random unoccupied positions (independent nuclei), and the remaining $N_{\text{new}} - N_{\text{seeds}}$ are grown by 6-neighbour (face-sharing) adjacency from the combined frontier of all occupied sites. Connectivity is guaranteed by construction: every occupied site is reachable from a seed through adjacency steps. The limits are transparent: $\kappa = 0$ places every new site as an independent random seed (Bernoulli site percolation); $\kappa = 1$ plants a single seed and grows a compact connected cluster (Eden growth [22]). Intermediate κ interpolates between a spatially uniform obstacle field and a clustered, compact network. The seed fraction $1 - \kappa$ is the nucleation density—the number of independent nuclei per newly occupied site—so κ indexes that density inversely.

RANDOM-WALK ENGINE AND OBSERVABLES

Random walkers are initialised on unoccupied sites, modelling microrheology probe particles in the fluid phase. Each step attempts a move to one of the six nearest neighbours chosen uniformly; if the target is unoccupied the move is accepted, otherwise the walker dwells at its current site for a binding time $\text{WT} = \lceil |\mathcal{N}(\mu_{\text{WT}}, \sigma_{\text{WT}})| \rceil$ steps, with $\mu_{\text{WT}} = 20$, $\sigma_{\text{WT}} = 5$. This models transient probe–network binding and suppresses artificial super-diffusion from immediate reflection. Periodic boundaries are applied through $x_n = \text{mod}(x_n - 1, L) + 1$; the unwrapped coordinate is used for the MSD,

$$\langle r^2(t) \rangle = \frac{1}{N_W} \sum_{i=1}^{N_W} [\Delta x_i(t)^2 + \Delta y_i(t)^2 + \Delta z_i(t)^2], \quad (\text{S2})$$

accumulated online by parallel reduction to avoid storing $O(L_W N_W)$ trajectories (~ 36 GB at production scale). Production runs use $N_W = 3000$ walkers and $N_s = 3$ replicates per condition. The anomalous exponent α is the log–log slope of $\langle r^2(t) \rangle$ over the intermediate window $[L_W/100, L_W/10]$, and the gel point is located by linear interpolation of $\alpha(p)$ through 0.5,

$$p'_c = p_i + \frac{0.5 - \alpha(p_i)}{\alpha(p_{i+1}) - \alpha(p_i)}(p_{i+1} - p_i), \quad \alpha(p_i) > 0.5 > \alpha(p_{i+1}). \quad (\text{S3})$$

WINTER-CHAMBON CRITERION AND THE GSER

At the gel point $\tan \delta = G''/G' = 1$, i.e. $\delta = 45^\circ$ [1, 2]. For a power-law MSD $\langle r^2(t) \rangle \propto t^\alpha$ the GSER [3, 4]

$$G^*(\omega) \approx \frac{k_B T}{\pi a \langle r^2(1/\omega) \rangle \Gamma(1 + \alpha(\omega))} \quad (\text{S4})$$

gives moduli $G', G'' \propto \omega^\alpha$ and a frequency-independent phase angle

$$\delta = \frac{\pi}{2} \alpha \iff \delta(^{\circ}) = 90 \alpha, \quad (\text{S5})$$

so $\delta = 45^\circ$ is exactly $\alpha = 0.5$. Dimensional spectra use $T = 293$ K, $\eta = 1.2 \times 10^{-3}$ Pa s, and probe radius $a = 0.243$ μm .

FINITE-SIZE-SCALING PROTOCOL AND RESULTS

We repeat the analysis at $L \in \{50, 100, 200, 300, 500, 750\}$ for $\kappa \in \{0, 0.6, 0.8, 0.97, 1.0\}$, supplemented by two crossover densities $\kappa \in \{0.99, 0.995\}$ run to $L \leq 500$ (the Eden-region lattice rebuild at $L = 750$ is prohibitively slow). Because the finite-size crossover time grows as L^2 , the steps per walker are scaled as $L_W(L) = 10^6 (L/500)^2$, keeping the fit window within the anomalous regime at every size. From the size-dependent gel points $p'_c(L)$ (Table S1) and logistic transition widths $w(L)$ we estimate the correlation-length exponent three ways: (i) the threshold shift $p'_c(L) = p'_c(\infty) + aL^{-1/\nu}$; (ii) the width scaling $w(L) \sim L^{-1/\nu}$; and (iii) the data collapse of $\alpha(p, L)$ against $(p - p'_c(\infty))L^{1/\nu}$, scored by the residual to a master curve in α -space (bounded, so axis rescaling cannot spuriously improve the score). The two smallest sizes ($L = 50, 100$) carry the strongest corrections to scaling and are excluded from all exponent fits; uncertainties are by jackknife over the fitted sizes.

Because $p'_c(L)$ is already stationary for $L \geq 200$ (Table S1), the shift fit determines $p'_c(\infty)$ robustly but has little leverage on ν ; the exponent is therefore taken from the width and collapse estimators, which do have leverage. These agree on

$$\nu_{\text{eff}} \approx 1.6, \quad \text{independent of } \kappa \text{ for } \kappa \leq 0.8, \quad (\text{S6})$$

as summarised in Table S3. The value differs from the static 3D site-percolation exponent $\nu = 0.88$ because p'_c is defined dynamically (through the probe-diffusion exponent) rather than through a static order parameter. Beyond $\kappa \approx 0.9$ the transition enters a crossover to the Eden singular limit: the gel point keeps rising smoothly— $p'_c(\infty) = 0.80, 0.857, 0.866$ for $\kappa = 0.97, 0.99, 0.995$ (Table S2), tracking monotonically towards the single-seed bound $p'_c \rightarrow 1$ —but the exponent is no longer resolved. With the crossover conditions capped at $L \leq 500$ (three fit sizes), the width and shift estimators become degenerate (e.g. $\nu_{\text{width}} = 68 \pm 44$ at $\kappa = 0.99$) and the collapse estimator, though finite, drifts to $\nu \approx 2.5$. We therefore report $\nu_{\text{eff}} \approx 1.6$ only for the resolved regime $\kappa \leq 0.8$ and treat $\kappa \geq 0.97$ as an unresolved crossover, not a second exponent. The Eden limit $\kappa = 1$ has $p'_c \rightarrow 1$ and no finite- L crossing within the accessible window, and is treated as a bound.

TABLE S1. Size-dependent gel point $p'_c(L)$ and its extrapolation $p'_c(\infty)$ (shift fit, $L \geq 200$).

L	$\kappa = 0$	$\kappa = 0.6$	$\kappa = 0.8$	$\kappa = 0.97$
50	0.734	0.754	0.773	0.881
100	0.693	0.720	0.740	0.840
200	0.684	0.706	0.725	0.815
300	0.683	0.705	0.724	0.812
500	0.681	0.702	0.721	0.805
750	0.682	0.703	0.724	0.804
∞	0.681	0.702	0.722	≈ 0.80

ORIGIN OF THE GEL CRITERION AND GEOMETRIC CROSS-CHECK

At the void-percolation threshold the probe explores a critical, fractal void, so its diffusion is anomalous with $\langle r^2(t) \rangle \propto t^{2/d_w}$, where d_w is the walk dimension of the incipient cluster [23–26]. For three-dimensional lattice percolation $d_w = 3.88(3)$ [27], i.e. $2/d_w = 0.515(4)$, within one percent of the Winter–Chambon value $\alpha = 0.5$. The

TABLE S2. Crossover set: size-dependent gel point $p'_c(L)$ for the strongly clustered densities $\kappa = 0.99, 0.995$, capped at $L \leq 500$. The gel point rises monotonically towards the Eden bound, but with only three fit sizes the exponent is not resolved (Table S3).

L	$\kappa = 0.99$	$\kappa = 0.995$
50	0.925	0.954
100	0.900	0.936
200	0.874	0.911
300	0.863	0.899
500	0.859	0.888
750	—	—
∞	0.857	0.866

TABLE S3. Effective correlation-length exponent from data collapse and transition-width scaling ($L \geq 200$; jackknife uncertainties). The shift fit does not constrain ν (see text). The $\kappa \geq 0.97$ rows are unresolved crossover points (capped at $L \leq 500$ for $\kappa \geq 0.99$), excluded from the reported ν_{eff} ; “deg.” marks a degenerate few-size fit.

κ	ν (collapse)	ν (width)
0.0	1.62	1.77 ± 0.21
0.6	1.64	1.49 ± 0.05
0.8	1.67	1.69 ± 0.03
0.97	2.50	2.94 ± 0.47 (crossover)
0.99	2.49	deg. (crossover, $L \leq 500$)
0.995	2.48	deg. (crossover, $L \leq 500$)

coincidence of the gel criterion with the percolation threshold is thus the 3D walk exponent falling close to 0.5, and is specific to three dimensions: in two dimensions $d_w = 2.878(1)$ gives $2/d_w = 0.70$, far from 0.5. This fixes the random ($\kappa = 0$) anchor, where the void at $p = 1 - p_c$ is itself a critical Bernoulli configuration [28].

For $\kappa > 0$ the void is the complement of a correlated (partially Eden) occupied phase, and whether the same picture persists is an empirical question, which two independent measurements answer. First, the effective width exponent is κ -independent for $\kappa \leq 0.8$ ($\nu_{\text{eff}} \approx 1.6$), distinct from the static $\nu = 0.88$ because p'_c is defined through the probe-diffusion exponent rather than a static order parameter. Second, the dynamical gel point p'_c^{MR} coincides with the purely geometric void-spanning threshold p'_c^{geom} for $\kappa \leq 0.8$ and decouples only in the Eden crossover (Fig. 4). Together these place the low- κ regime in the ordinary 3D percolation class and identify $\kappa \rightarrow 1$ as a distinct compact-solidification limit: the void of a single compact cluster retains a spanning path—by translational (periodic) invariance, independent of the cluster’s position—until $p \rightarrow 1$, so $p'_c(\kappa=1) = 1$ marks the point at which the last fluid vanishes rather than an extrapolated bound.

Geometric threshold.—On the same lattices (identical κ growth, seeds and cumulative construction), we label the unoccupied sublattice with 6-connectivity (Hoshen–Kopelman) and test whether one void cluster touches both faces along an axis. Averaging the spanning probability over $N_s = 3$ seeds at $L = 500$ and locating its 0.5 crossing gives $p'_c^{\text{geom}}(\kappa)$; at $\kappa = 0$ this returns $0.685 \approx 1 - p_c$, validating the estimator, and face-to-face and periodic-wrapping definitions agree in the thermodynamic limit. The dynamical threshold lies slightly above the geometric one for $\kappa \gtrsim 0.85$ because, once the void fragments into large finite pockets, an ensemble of probes still records $\alpha > 0.5$ over the fit window before feeling the pocket walls; passive microrheology therefore mildly overestimates structural gelation in the compact regime.

FIXED- L GEL-POINT CURVE $p'_c(\kappa)$

At $L = 500$, an 18-value κ -grid resolves the full gel-point curve (Table S4; mean over $N_s = 3$ replicates, $\alpha = 0.5$ crossing). The curve is monotonic and consistent with the thermodynamic-limit anchors of Table S1. Two regimes are separated near a $\sim 1\%$ seed fraction ($\kappa \approx 0.99$): a gradual rise for $\kappa \lesssim 0.99$ (many nuclei, quasi-random morphology) and a steep rise towards unity for $\kappa \gtrsim 0.99$ (single-cluster Eden morphology).

TABLE S4. Gel point p'_c at $L = 500$ for all 18 nucleation densities.

κ	$p'_c(L=500)$	κ	$p'_c(L=500)$
0.00	0.683	0.94	0.773
0.20	0.687	0.97	0.806
0.40	0.692	0.98	0.824
0.60	0.703	0.99	0.852
0.70	0.709	0.995	0.880
0.80	0.721	0.996	0.888
0.85	0.733	0.997	0.902
0.88	0.744	0.998	0.910
0.91	0.751	1.00	$\rightarrow 1$ (Eden bound)

FREQUENCY-DOMAIN (GSER) SPECTRA

MSD curves near p'_c are converted to $G'(\omega)$, $G''(\omega)$, and $\delta(\omega)$ via the GSER. Because the MSD is close to a single power law over the anomalous window, the spectra are power laws with frequency-independent $\delta = 90\alpha$ over the accessible range $\omega \approx 0.4\text{--}4 \text{ rad s}^{-1}$. At the gel point $G' \approx G''$ and $\delta \approx 45^\circ$ for every κ (Fig. S2); below it $\delta > 45^\circ$ ($G'' > G'$, viscous), above it $\delta < 45^\circ$ ($G' > G''$, elastic). The current spectra are single-replicate per condition; replication would tighten the modulus estimates.

ROBUSTNESS: ALTERNATIVE GROWTH RULES

Connectivity-preserving growth from many independently seeded nuclei per step (an adjacency-templated rule) also elevates the gel point; for the specific rule studied, $p'_c(\infty) \approx 0.93$. Six-neighbour and twenty-six-neighbour adjacency give indistinguishable gel points, showing the shift is driven by the connectivity constraint itself, not by the precise adjacency rule. This shows that morphology control of the gel point is robust across the growth rules tested, rather than an artefact of the κ construction; the distinct p'_c of this rule simply reflects a different point in growth-morphology space.

COMPUTATIONAL IMPLEMENTATION AND AVAILABILITY

Simulations were implemented in MATLAB with `parfor` parallelisation over walkers. Lattices are generated cumulatively as described above; the random-walk engine tracks both wrapped coordinates (for neighbour lookup) and unwrapped coordinates (for the MSD), and the ensemble MSD is accumulated online by parallel reduction. The finite-size-scaling driver runs each κ condition at all six sizes with $L_W = 10^6(L/500)^2$, $N_W = 3000$, $N_s = 3$, and the analysis extracts $p'_c(L)$, $w(L)$, and the three exponent estimates with jackknife uncertainties, fits restricted to $L \geq 200$. Code and processed data are available from the authors on request [or: at [repository link](#)].

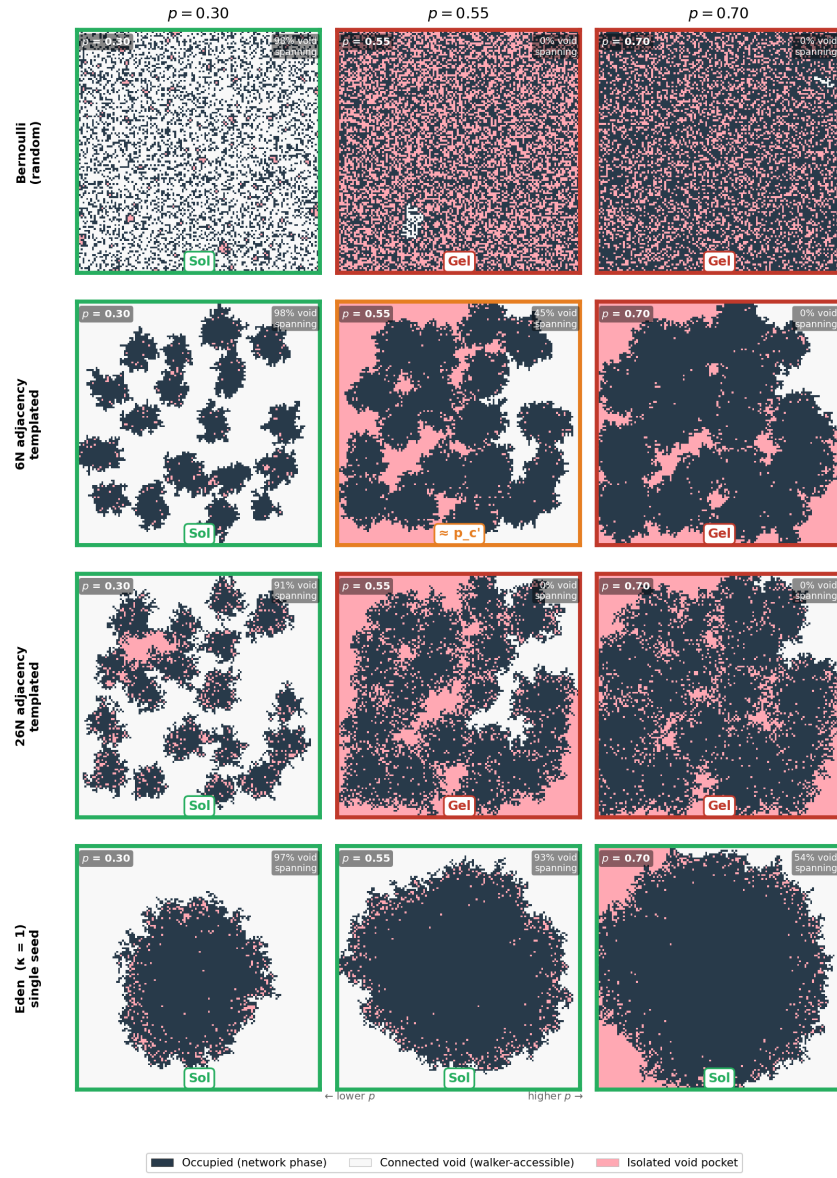


FIG. S1. Schematic of growth morphologies at fixed occupation, from random ($\kappa = 0$) through clustered intermediate κ to single-seed (Eden, $\kappa = 1$) growth, illustrating how compact growth leaves a more connected void and thus a higher gel point.

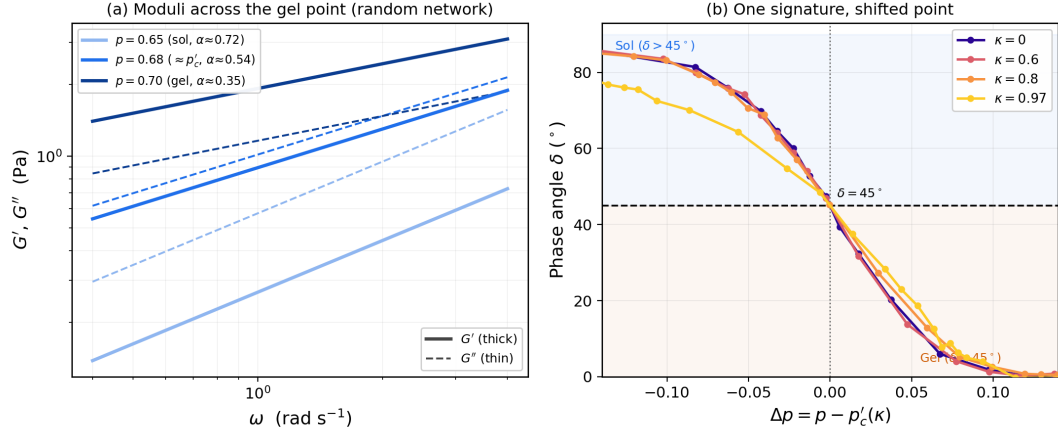


FIG. S2. **One signature, shifted point.** (a) GSER-derived $G'(\omega)$, $G''(\omega)$ for the random ($\kappa = 0$) network at occupation densities straddling p'_c : viscous ($G'' > G'$) below, elastic ($G' > G''$) above, and $G' \approx G''$ ($\delta \approx 45^\circ$) at the gel point—the Winter–Chambon signature. (b) Phase angle $\delta = 90\alpha$ versus distance from the gel point $\Delta p = p - p'_c(\kappa)$ for several κ ; all collapse through $\delta = 45^\circ$ at $\Delta p = 0$ —the same critical rheology at each κ 's own (shifted) gel point.